

## **Program 6: Deploy an automated build pipeline using docker hub**

File structure:

/Dockerfile

/package.json

Create folder >> mkdir lab6 >> cd lab6

### **Dockerfile**

FROM alpine

Using Docker File

```
#sudo docker build -t program6 .
```

Create github personal access token

Login to github > settings > developer settings > personal access tokens > classic > give name and expiry date > check write packages and read packages

Create and copy the pat

Tag image for dockerhub

```
#docker tag program6 gaytri3011 program6:latest
```

Tag image for ghcr.io

```
#docker tag program6 ghcr.io/anantham016/program6:latest
```

Push to the docker hub

```
#docker login -u gaytri3011 #docker  
push gaytri3011/program6
```

Login to GHCR using docker

```
#echo <PAT>| docker login ghcr.io -u gaytri3011--password-stdin
```

Push the image to ghcr

```
#docker push ghcr.io/gaytri3011/program6
```

```
1RV24MC036_gayatri_v_k@ubuntu:~$ mkdir program6
1RV24MC036_gayatri_v_k@ubuntu:~$ cd program6
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ nano index.js
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ npm init -y
Wrote to /home/1RV24MC036_GAYATRI_V_K/program6/package.json:
```

```
{
  "name": "program6",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

```
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ npm install
```

```
up to date, audited 1 package in 232ms
```

```
found 0 vulnerabilities
```

```
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ ls
```

```
index.js  package.json  package-lock.json
```

```
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ nano Dockerfile
```

```
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ docker login
```

```
Authenticating with existing credentials... [Username: gaytri3011]
```

 **Info** → To login with a different account, run 'docker logout' followed by 'docker login'

```
Login Succeeded
```

```
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ docker pull node:18
```

```
18: Pulling from library/node
```

```
Digest: sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
```

```
Status: Image is up to date for node:18
```

```
docker.io/library/node:18
```

```
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ docker build -t program6 .
```

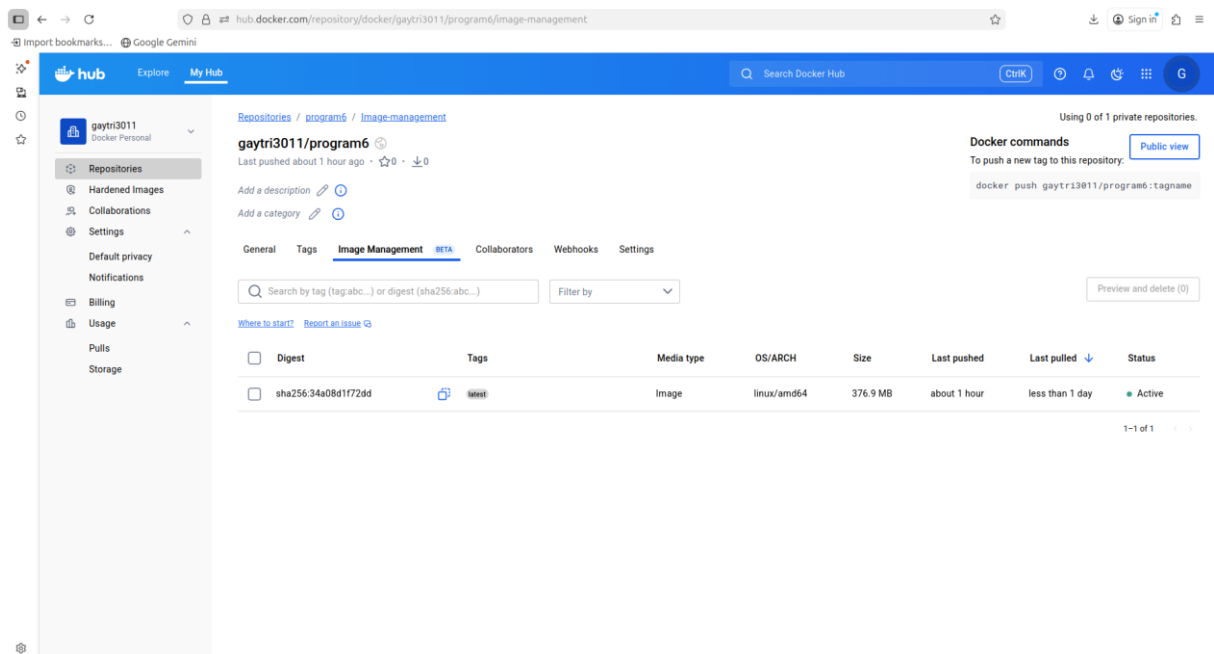
```
[+] Building 2.5s (10/10) FINISHED docker:default
```

```
=> [internal] load build definition from Dockerfile 0.0s
```

```
=> => transferring dockerfile: 167B 0.0s
```

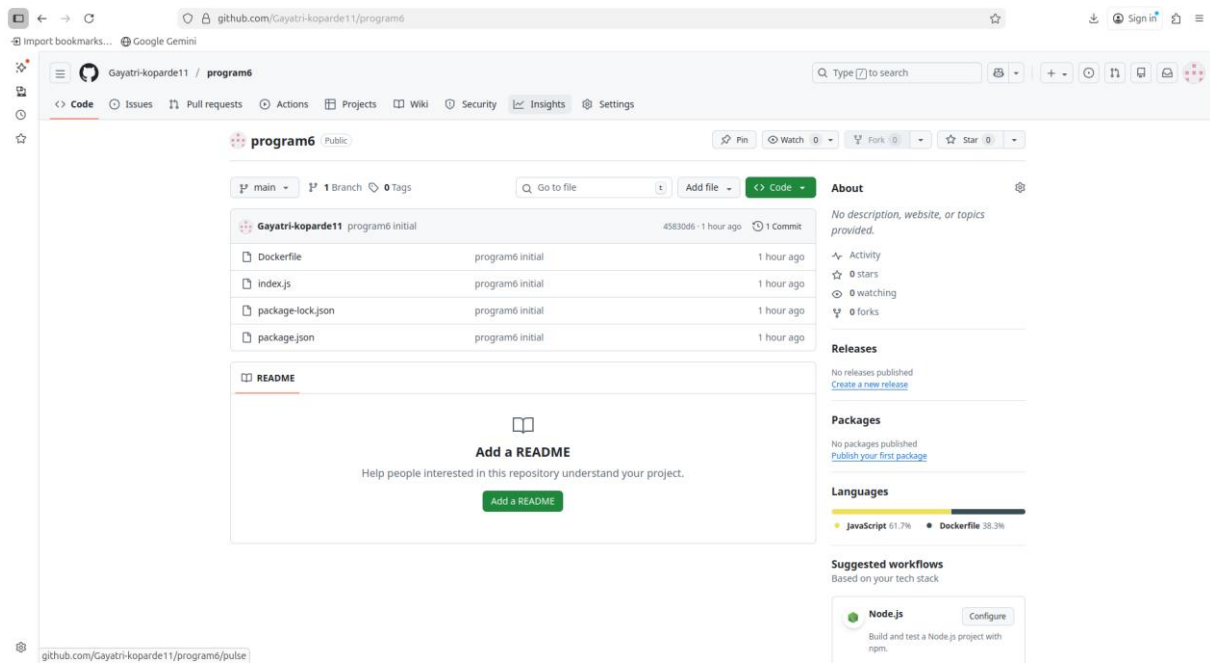
```
=> [internal] load metadata for docker.io/library/node:18 0.0s
```

```
=> [internal] load definition for docker.io/library/node:18 0.0s
```



The screenshot shows the Docker Hub interface for the repository `gaytri3011/program6`. The page is titled "Repositories / program6 / Image-management". It shows the repository details, including the last push time ("Last pushed about 1 hour ago") and the number of stars (0). The "Image Management" tab is selected, showing a table of images. The table has columns for Digest, Tags, Media type, OS/ARCH, Size, Last pushed, Last pulled, and Status. One image is listed with Digest `sha256:34a08d172dd` and Tag `test`. The status is "Active".

| Digest             | Tags | Media type | OS/ARCH     | Size     | Last pushed  | Last pulled     | Status |
|--------------------|------|------------|-------------|----------|--------------|-----------------|--------|
| sha256:34a08d172dd | test | Image      | linux/amd64 | 376.9 MB | about 1 hour | less than 1 day | Active |



Create a repository and add file to  
github #git init  
#git add .  
#git commit -m "first commit"  
#git branch main  
#git remote add origin <remote\_repository\_url>  
#git push -u origin Main

```
d399c9dc306f: Mounted from gaytri3011/programm6
84f9fa179c1b: Mounted from gaytri3011/programm6
ce84ba212e49: Mounted from gaytri3011/programm6
e4dc8cd9ecc8: Mounted from gaytri3011/programm6
6428cc293366: Mounted from gaytri3011/programm6
2f7436e79a0b: Mounted from gaytri3011/programm6
latest: digest: sha256:34a08d1f72ddb7a24baf60d9d11cf3fee1b174adb68414d0b7a0ac9462d126b size: 2831
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1RV24MC036_GAYATRI_V_K/program6/.git/
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ git branch -M main
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ git branch -M main
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ git remote add origin https://github.com/gayatri-koparde11/program6.git
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ git add .
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ git commit -m "program6 initial"
[master (root-commit) 45838d6] program6 initial
 4 files changed, 38 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 index.js
 create mode 100644 package-lock.json
 create mode 100644 package.json
1RV24MC036_gayatri_v_k@ubuntu:~/program6$ git push -u origin main
Username for 'https://github.com': gayatri-koparde11
Password for 'https://gayatri-koparde11@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 890 bytes | 445.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
remote: This repository moved. Please use the new location:
remote:   https://github.com/Gayatri-koparde11/program6.git
To https://github.com/gayatri-koparde11/program6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
1RV24MC036_gayatri_v_k@ubuntu:~/program6$
```